Ball lifts of the split Cayley hexagon: bipartite graphs of diameter 9, 10 and 11

Degree–diameter lab (Claude Code sessions coordinated by Doğa Özden)

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Draft of 28 October 2026 (job thm-polygon-levels; branch `claude/haiku-squads-diameter-7-hr64vz` of github.com/dogaozden/degree-diameter). Written in the style of the lab's diameter-7 note (research/theorems/D7_HEXAGON.md on branch `claude/d7-hexagon-theorems-4w1prt`, "the D7 note"), whose coordinates and notation are used. Every claim is labelled **Theorem** (proved here, by hand), **Computer-checked** (a statement about specific finite objects, with the check named) or **Heuristic** (an argument with a gap that is named).

0. Results

The diameter-7 level graphs (copies of the incidence graph of $H(q)$ joined by matchings, the 2013 table's $D = 7$ column) let a walk change layers only at its first or last vertex. Here the jump may start anywhere in a ball of radius ρ around the walk's end, and the layer it reaches depends on the **colour class** of the jumping element. This gives one theorem for all diameters $c + 2 + \rho$:

- **Theorem 2.3 (ball lift).** For any bipartite graph I with point–line distances $\leq c$, k balanced ρ-dominating colour classes on each side give a $\mathbb{Z}_{2rk+1}$-lift of I plus r matchings, of diameter $\leq \rho + c + 2$. The classical level graph is the case $\rho = 0$, $k = 1$.
- $D = 9$ **(Corollary 3.1):** $2N(q)(2rk+1)$ vertices, degree $q+1+r$, where k = number of balanced classes meeting every plane π_p (points) and every line 2-ball (lines). $k \leq q^2 + q + 1$, with equality iff $H(q)$ has a partition into ovoids and one into spreads (Proposition 3.2); $H(3)$ has no three pairwise disjoint spreads (Computer-checked 3.3). At $k = q^2 + q + 1$ the construction would beat the lab's value at every $(\Delta, 9)$, $7 \leq \Delta \leq 20$, except $\Delta = 8$; the record thresholds need k between 49% and 97% of $q^2 + q + 1$. DTHREE_PLACEHOLDER
- $D = 10$ **(Corollary 4.4):** for every q, a Hermitian spread gives $2N(q)(2r'(q^3+1)+1)$ vertices at degree $q + 1 + 2r'$, with no search. It reaches 84–97% of the lab's values at $(16, 10)$, $(19, 10)$, $(20, 10)$ and beats none.
- $D = 11$ **(Corollaries 5.2, 5.5):** K pairwise disjoint totally singular lines of $Q(6, q)$ (any line of the quadric is 4-dominating, Lemma 5.4) and K pairwise non-collinear points of $H(q)$ give $2N(q)(2rK+1)$ vertices at degree $q+1+r$; for $q = 3^h$ the duality makes the second condition automatic. DELEVEN_PLACEHOLDER
- $D = 8$**, quadrangles, the Ree–Tits octagon (§6):** no cells.

All theorems were tested by building the graphs at small q and running an all-pairs BFS (§8).

1. Notation

Throughout, q is a prime power and $H = H(q)$ is the split Cayley hexagon in the coordinates of the D7 note, §1.1: $V = \mathbb{F}_q^7$ with coordinates $(X_0, \ldots, X_6)$, the parabolic quadric

$$Q(X) = X_0X_4 + X_1X_5 + X_2X_6 - X_3^2,$$

with polar form $b(X,Y) = Q(X+Y) - Q(X) - Q(Y)$. The points of H are the N points of $Q(6,q)$,

$$N = N(q) = \frac{q^6-1}{q-1} = (q+1)(q^4+q^2+1) = (q^3+1)(q^2+q+1),$$

the lines are the N lines of $Q(6,q)$ whose Grassmann coordinates satisfy Tits' equations $p_{12} = p_{34}$, $p_{54} = p_{32}$, $p_{20} = p_{35}$, $p_{65} = p_{30}$, $p_{01} = p_{36}$, $p_{46} = p_{31}$, and incidence is containment. P and L denote the point and line sets.

$I = I(q)$ is the incidence graph of H (vertex set $P \cup L$, $2N$ vertices, $(q+1)$-regular, bipartite), d its distance. H is a generalized hexagon of order (q,q), so (D7 note §1.2):

- (F1) I has diameter 6 and girth 12; point–line distances are $1, 3, 5$ and same-type distances $0, 2, 4, 6$.
- (F2) if $d(x,y) \le 5$ there is a unique shortest path from x to y.

For a point x, the lines of H through x are the $q+1$ lines through x of a totally singular plane π_x of $Q(6,q)$, and the points at distance ≤ 2 from x in I are exactly the q^2+q+1 points of π_x (each line through x contributes its q further points).

Balls. For an element x and an even $\rho \ge 0$, $B_\rho(x)$ is the set of elements **of the same type** as x at distance $\le \rho$ from x. So $B_0(x) = \{x\}$; $B_2(p) = \pi_p$ for a point p; $B_2(l)$ is l together with the $q(q+1)$ lines meeting l; and $|B_2(x)| = q^2+q+1$ for every x. A set A of elements of one type is **ρ-dominating** if it meets $B_\rho(x)$ for every element x of that type.

Ovoids and spreads. An ovoid of H is a set of q^3+1 pairwise opposite points (distance 6); a spread is a set of q^3+1 pairwise opposite lines. Every element is at distance ≤ 2 from exactly one element of an ovoid (resp. spread) of its type: two elements of an ovoid cannot share a 2-ball (they would be at distance ≤ 4), and $(q^3+1)(q^2+q+1) = N$. Spreads exist for every q (the Hermitian spread: the lines of H contained in a hyperplane section $Q^-(5,q)$); ovoids exist when q is a power of 3 (then $H \cong H^D$ and a duality maps a spread to an ovoid); see §9 for non-existence results.

2. The ball lift

The construction works for any bipartite graph in place of I; the hexagon enters only through the numbers in Corollaries 3.1, 4.4, 5.2 and 5.5. Fix:

- a connected bipartite graph I with colour classes P ("points") and L ("lines"), $|P| = |L| = N$, in which every point–line distance is at most an odd number c;
- an even $\rho \ge 0$ and integers $k, r \ge 1$;
- partitions $P = A_0 \cup \cdots \cup A_{k-1}$ and $L = B_0 \cup \cdots \cup B_{k-1}$ with $|A_i| = |B_i|$, such that every A_i is ρ-dominating (meets $B_\rho(p)$ for every point p) and every B_i is ρ-dominating (meets $B_\rho(l)$ for every line l);
- bijections $\sigma_1, \dots, \sigma_r : P \to L$ with $\sigma_t(A_i) = B_i$ for all t, i.

Put $m = 2rk+1$ and, for $z \in A_i$ and $1 \le t \le r$, $e_t(z) = (t-1)k + i + 1 \in \{1, \dots, rk\}$.

Definition 2.1 (the ball lift $\Lambda = \Lambda(I; A, B, \sigma; r)$). The vertex set is $(P \cup L) \times \mathbb{Z}_m$. The neighbours of a point vertex (z,g), $z \in A_i$, are

$$(l,g) \ \ (l \ni z), \qquad (\sigma_t(z),\ g + (t-1)k + i + 1) \ \ (1 \le t \le r);$$

the neighbours of a line vertex (l,g), $l \in B_i$, are

$$(p,g) \ \ (p \in l), \qquad (\sigma_t^{-1}(l),\ g - (t-1)k - i - 1) \ \ (1 \le t \le r)$$

(note $\sigma_t^{-1}(l) \in A_i$, so the two lists describe the same edges). In words: m copies ("layers") of I, and r perfect matchings between points and lines, the edge at z of the t-th matching climbing $e_t(z)$ layers.

Lemma 2.2 (Theorem). Let Γ be a connected bipartite graph with at least one edge. (a) If any two vertices of the same colour are at distance $\le 2s$, then $\operatorname{diam}\Gamma \le 2s+1$. (b) If any two vertices of different colours are at distance $\le 2s-1$, then $\operatorname{diam}\Gamma \le 2s$.

Proof. (a) Let u, v have different colours and let v' be a neighbour of v (it exists: Γ is connected with an edge). v' has the colour of u, so $d(u,v) \le d(u,v') + 1 \le 2s+1$. (b) The same with u, v of the same colour. $\square$

Theorem 2.3. Λ is a simple bipartite graph on $2Nm$ vertices in which every vertex has degree $\deg_I(x) + r$, and

$$\operatorname{diam}\Lambda \;\le\; \rho + c + 2.$$

Proof. Simple and bipartite. Every edge joins $P \times \mathbb{Z}_m$ to $L \times \mathbb{Z}_m$. An edge of I keeps the layer; the t-th matching edge at (z,g) changes it by $e_t(z) \in [(t-1)k+1,\, tk]$, which is never 0 modulo $m = 2rk+1$, and differs for different t. So no edge is repeated, and each vertex has $\deg_I(x)$ edges inside its layer and r matching edges.

Translations. $\tau_h : (x,g) \mapsto (x, g+h)$ is an automorphism, so it suffices to bound distances from vertices in layer 0. Write $F = \{1, \dots, rk\}$; then $\mathbb{Z}_m = \{0\} \cup F \cup (-F)$, a disjoint union, and every $h \in F$ is $(t-1)k + i + 1$ for exactly one pair (t,i).

Two points. Let $p, p' \in P$ and $h \in \mathbb{Z}_m$; we show $d((p,0),(p',h)) \le \rho + c + 1$.

- $h = 0$: inside layer 0, $d_I(p,p') \le c+1$ (go from p to a line through p', then to p').
- $h = (t-1)k + i + 1 \in F$: choose $z \in A_i \cap B_\rho(p)$. The walk $(p,0) \rightsquigarrow (z,0) \to (\sigma_t(z), h) \rightsquigarrow (p',h)$ has length $\le \rho + 1 + c$ (the middle edge is the t-th matching edge at z, which climbs $e_t(z) = h$ layers; the last segment is a point–line path of I inside layer h).
- $h = -((t-1)k + i + 1) \in -F$: choose $z \in A_i \cap B_\rho(p')$. The walk $(p,0) \rightsquigarrow (\sigma_t(z), 0) \to (z,h) \rightsquigarrow (p',h)$ has length $\le c + 1 + \rho$.

Two lines. Let $l, l' \in L$.

- $h = 0$: $\le c+1$ inside layer 0.
- $h = -((t-1)k + i + 1)$: choose $w \in B_i \cap B_\rho(l)$; then $\sigma_t^{-1}(w) \in A_i$ and the walk $(l,0) \rightsquigarrow (w,0) \to (\sigma_t^{-1}(w), h) \rightsquigarrow (l',h)$ has length $\le \rho + 1 + c$.
- $h = (t-1)k + i + 1$: choose $w \in B_i \cap B_\rho(l')$; the walk $(l,0) \rightsquigarrow (\sigma_t^{-1}(w), 0) \to (w,h) \rightsquigarrow (l',h)$ has length $\le c + 1 + \rho$.

So any two vertices of the same colour are within $\rho + c + 1$ (an even number). In particular Λ is connected (every vertex has a neighbour), and Lemma 2.2(a) gives $\operatorname{diam}\Lambda \le \rho + c + 2$. $\square$

Remark 2.4 (the classical case). $\rho = 0$, $k = 1$, $A_0 = P$, $B_0 = L$: the hypotheses hold trivially ($B_0(x) = \{x\}$) and Λ is the level graph with $2r+1$ layers and arbitrary matchings, of diameter $\le c+2$. For $H(q)$ ($c = 5$) this is Theorem A of the D7 note (the 2013 table's diameter-7 values $(2r+1)\cdot 2N(q)$); for a generalized quadrangle ($c = 3$) it is the diameter-5 polygon cover. The point of Theorem 2.3 is that a jump may start anywhere in a ball of radius ρ, so ρ more steps of diameter buy a factor $\approx |B_\rho|$ in the number of layers instead of a factor $\approx r$.

Remark 2.5 (relaxed data). The partitions can be replaced by pairwise disjoint ρ-dominating families $D_0, \dots, D_{k-1} \subseteq P$ and $E_0, \dots, E_{k-1} \subseteq L$ with $\sum_i \max(|D_i|, |E_i|) \le N$: enlarge the smaller of D_i, E_i by unused elements until $|D_i| = |E_i|$, then distribute the rest evenly (enlarging a dominating set keeps it dominating). This is the balance condition checked by the scorer `research/theorems/polylev/domcheck.py`.

Remark 2.6 (dualities). If φ is a duality of H (an incidence-preserving bijection $P \cup L \to L \cup P$ exchanging the types), then φ maps $B_\rho(p)$ onto $B_\rho(\varphi(p))$, so $\varphi(A_i)$ is a ρ-dominating set of lines with $|\varphi(A_i)| = |A_i|$. Hence when q is a power of 3, a point partition alone suffices: take $B_i = \varphi(A_i)$ and $\sigma_t = \varphi|_P$ (or any bijection with $\sigma_t(A_i) = B_i$).

3. Diameter 9 from $H(q)$

Corollary 3.1 (Theorem). Let $A_0, \dots, A_{k-1}$ be a partition of the points of $H(q)$ in which every A_i meets every plane π_p ($p \in P$), and $B_0, \dots, B_{k-1}$ a partition of the lines in which every B_i contains, for every line l, the line l or a line meeting l; suppose $|A_i| = |B_i|$. Then for every $r \ge 1$ there is a bipartite $(q+1+r)$-regular graph of diameter at most 9 with

$$2N(q)\,(2rk+1) \quad \text{vertices.}$$

The graph is Λ of Definition 2.1 with $I = I(q)$ (so $c = 5$), $\rho = 2$ and any bijections σ_t with $\sigma_t(A_i) = B_i$; Theorem 2.3 gives the diameter. Explicitly, the neighbours of (z, g), $z \in A_i$, are the $q + 1$ vertices (l, g) with $l \ni z$ and the r vertices $(\sigma_t(z), g + (t-1)k + i + 1)$; the neighbours of (l, g), $l \in B_i$, are the $q + 1$ vertices (p, g) with $p \in l$ and the r vertices $(\sigma_t^{-1}(l), g - (t-1)k - i - 1)$.

Proposition 3.2 (Theorem). In Corollary 3.1, $k \le q^2 + q + 1$, with equality if and only if every A_i is an ovoid and every B_i is a spread. Hence $k = q^2 + q + 1$ needs $H(q)$ to have ovoids (they exist when q is a power of 3; $H(2)$ has none, Computer-checked 3.3; see §9 for what is known in general).

Proof. A point z lies in π_p exactly for the q^2+q+1 points $p \in \pi_z$. Since A_i meets all N planes, $|A_i|(q^2+q+1) \ge N$, i.e. $|A_i| \ge q^3 + 1$, and $\sum_i |A_i| = N$ gives $k \le q^2 + q + 1$. Equality forces $|A_i| = q^3 + 1$ and every plane to meet A_i exactly once; then two points of A_i are never collinear (both would lie in π_z for one of them) and never at distance 4 (both would lie in π_y for the middle point y of the path), so A_i is an ovoid. The same for lines. Conversely, ovoids and spreads meet every 2-ball exactly once (§1). □

Computer-checked 3.3 (research/theorems/polylev/perfcodes.c, triples.py). $H(2)$ has no ovoid and exactly 28 spreads. $H(3)$ has exactly 6,318 spreads and 6,318 ovoids (exact-cover enumeration of the perfect codes of the 2-ball hypergraph); there are 3,411,720 disjoint pairs of spreads but **no three pairwise disjoint spreads**, so in particular no partition into 13 spreads, and $k = 13$ is impossible at $q = 3$. (The search for a partition was also run directly as an exact cover and found none.)

So the extremal case of Corollary 3.1 does not occur at $q = 3$, and near-extremal partitions are rare: dominating sets only slightly larger than ovoids pack badly. Partitions with k well below $q^2 + q + 1$ do exist:

Computer-checked 3.4 (local search, research/theorems/polylev/domlocal.c and translocal.c; every colouring was re-checked by the independent scorer domcheck.py, which tests every 2-ball).

q	side	k found	$q^2 + q + 1$	method

DOMTABLE_PLACEHOLDER

Sizes (research/theorems/polylev/sizes.py). With k at its maximum $q^2 + q + 1$, and the smallest k that beats the lab's current value at the same $(\Delta, 9)$ (rows where even the maximum does not are omitted):

Delta	q	r	max k	order at max k	lab value	k needed to beat it
7	5	1	31	492,156	462,840	30 (97%)
9	7	1	57	4,509,840	3,499,776	45 (79%)
10	7	2	57	8,980,464	8,003,856	51 (89%)
10	8	1	73	11,010,006	8,003,856	53 (73%)
11	8	2	73	21,945,114	16,792,080	56 (77%)
11	9	1	91	24,313,380	16,792,080	63 (69%)
12	9	2	91	48,493,900	34,026,720	64 (70%)
13	9	3	91	72,674,420	67,742,688	85 (93%)
13	11	1	133	94,601,304	67,742,688	96 (72%)
14	11	2	133	188,848,296	116,068,656	82 (62%)
15	11	3	133	283,095,288	208,029,120	98 (74%)
15	13	1	183	295,239,756	208,029,120	129 (70%)
16	11	4	133	377,342,280	337,394,400	119 (89%)
16	13	2	183	589,675,044	337,394,400	105 (57%)
17	13	3	183	884,110,332	501,707,400	104 (57%)
18	13	4	183	1,178,545,620	790,626,816	123 (67%)
18	16	1	273	1,223,618,214	790,626,816	177 (65%)
19	13	5	183	1,472,980,908	1,218,394,320	152 (83%)
19	16	2	273	2,444,999,466	1,218,394,320	136 (50%)
19	17	1	307	1,855,575,540	1,218,394,320	202 (66%)

Delta	q	r	max k	order at max k	lab value	k needed to beat it
20	16	3	273	3,666,380,718	1,823,175,864	136 (50%)
20	17	2	307	3,708,133,884	1,823,175,864	151 (49%)

3.5 Group-equivariant colourings. Let $T = \langle h \rangle$ be a cyclic group of collineations of H of order d acting semiregularly on points and on lines, and fix a representative b_O of every T-orbit O. Given a block $a_O \in \{0, \dots, A-1\}$ and a shift $s_O \in \mathbb{Z}_d$ for every orbit, colour $h^j b_O$ by $a_O d + (j + s_O \bmod d)$. Then h maps the class of colour (a, c) onto the class $(a, c+1)$ and maps 2-balls to 2-balls, so a 2-ball $B_2(h^j x)$ sees every colour if and only if $B_2(x)$ does: it suffices to check one 2-ball per orbit, which shrinks the search by a factor d. If the numbers of orbits in the A blocks are the same on the point side and on the line side (the `fixed` mode of translocal.c), every colour class has as many points as lines, which is the balance condition of Corollary 3.1; and σ_t can be taken T-equivariant. We found (g2.py: linear collineations obtained by mapping the coordinate apartment onto random apartments, products of two of them, powers):

- **Computer-checked 3.6.** Cyclic groups of order $q^2 - q + 1$ acting semiregularly on the points and lines of $H(q)$ for $q = 3, 4, 7, 9$ (orders 7, 13, 43, 73; grp/T_q*.h, the 7×7 matrices; every line is mapped to a line). For $q = 5$ the group of order 21 has line orbits of size 7 (an element of order 3 fixes lines; $3 \mid q+1$), so only its subgroup of order 7 is semiregular; the same happens for every $q \equiv 2 \pmod 3$.
- The Singer cycles of $SL(3, q) \le G_2(q)$, acting by $(v, a, w) \mapsto (gv, a, g^{-T} w)$ in the coordinates $x = (v, a, w)$, $v = (x_0, x_1, x_2)$, $a = x_3$, $w = (x_4, x_5, x_6)$, have order $q^2 + q + 1$; they act semiregularly when $q \not\equiv 1 \pmod 3$ (a fixed point would need an eigenvector of g over $\mathbb{F}_q$), and their subgroups of order prime to 3 act semiregularly for all q (grp/S*_q*.pperm: orders 7 and 13 at $q = 9$, 7 and 19 at $q = 11$, 61 at $q = 13$, 91 at $q = 16$).

4. Diameter 10: a one-sided ball lift and spread stars

For even diameter only one side needs colouring classes.

Definition 4.1. Let I, P, L, N, c be as in §2, let $\rho' \ge 1$ be odd, and let $L = C_0 \cup \dots \cup C_{K-1}$ be a partition of the lines such that every C_i meets $\{l \in L : d(p, l) \le \rho'\}$ for every point p. Let $r' \ge 1$, $m = 2r'K + 1$, and let $\sigma_{t,+}, \sigma_{t,-} : P \to L$ $(1 \le t \le r')$ be **arbitrary** bijections. Write $\iota(l) = i$ for $l \in C_i$. The graph Λ' has vertex set $(P \cup L) \times \mathbb{Z}_m$; the neighbours of (z, g), $z \in P$, are

$$(l, g)\ (l \ni z), \quad (\sigma_{t,+}(z),\ g + (t-1)K + \iota(\sigma_{t,+}(z)) + 1), \quad (\sigma_{t,-}(z),\ g - (t-1)K - \iota(\sigma_{t,-}(z)) - 1) \qquad (1 \le t \le r'),$$

and the neighbours of (l, g) are (p, g) $(p \in l)$, $(\sigma_{t,+}^{-1}(l), g - (t-1)K - \iota(l) - 1)$ and $(\sigma_{t,-}^{-1}(l), g + (t-1)K + \iota(l) + 1)$.

Theorem 4.2. Λ' is a simple bipartite graph on $2Nm$ vertices, each vertex has degree $\deg_I + 2r'$, and $\operatorname{diam} \Lambda' \le \rho' + c + 2$.

Proof. The layer changes of the $2r'$ matching edges at a point are pairwise distinct and nonzero modulo m (they lie in $\pm[(t-1)K + 1, tK]$), so Λ' is simple; it is bipartite with classes $P \times \mathbb{Z}_m$, $L \times \mathbb{Z}_m$. By the translations it suffices to bound $d((p, 0), (l', h))$ for a point p and a line l'. If $h = 0$ this is $\le c$. If $h = -((t-1)K + i + 1)$, pick $l \in C_i$ with $d(p, l) \le \rho'$; then $(p, 0) \rightsquigarrow (l, 0) \to (\sigma_{t,+}^{-1}(l), h) \rightsquigarrow (l', h)$ has length $\le \rho' + 1 + c$. If $h = (t-1)K + i + 1$, do the same with $\sigma_{t,-}$. So every point–line distance is $\le \rho' + c + 1$ (odd), and Lemma 2.2(b) gives the claim. □

Lemma 4.3 (Theorem). Let S be a spread of $H(q)$ and, for $s \in S$, let $C_s = B_2(s)$ (the line s and the lines meeting it). The C_s partition L, and every C_s contains a line at distance ≤ 3 from any given point.

Proof. The partition is the perfect-code property of spreads (§1). Let p be a point and $s \in S$; $d(p, s) \in \{1, 3, 5\}$. If $d(p, s) = 1$, take s. If $d(p, s) = 3$, the path $p\, m\, x\, s$ gives the line $m \ni p$ meeting s, at distance 1. If $d(p, s) = 5$, the path $p\, m_1\, x_1\, m_2\, x_2\, s$ gives $m_2 \in C_s$ at distance 3 from p. □

Corollary 4.4 (Theorem). For every prime power q and $r' \ge 1$ there is a bipartite $(q + 1 + 2r')$-regular graph of diameter at most 10 with $2N(q)\,(2r'(q^3 + 1) + 1)$ vertices (Theorem 4.2 with $I = I(q)$, $c = 5$, $\rho' = 3$, and the classes C_s of a Hermitian spread, $K = q^3 + 1$).

Computer-checked 4.5 (squad card r1/c14, research/squads/polylev/r1/c14/build_star.py; apsp.c). The graphs of Corollary 4.4 with $r' = 1$ and random bijections were built for $q = 2$ (2,394 vertices, degree 5) and $q = 3$ (41,496 vertices, degree 6), two seeds each; all-pairs BFS: bipartite, regular, diameter 8 ($q = 2$) and 9 ($q = 3$), within the bound 10.

The orders against the lab's current values at $D = 10$ (research/theorems/polylev/sizes.py):

Delta	q	matchings 2r'	K = q^3+1	order	lab value	ratio
5	2	2	9	2,394	95,472	0.025
6	3	2	28	41,496	524,160	0.079
7	2	4	9	4,662	2,298,240	0.002
7	4	2	65	357,630	2,298,240	0.156
8	3	4	28	82,264	8,043,840	0.010
8	5	2	126	1,976,436	8,043,840	0.246
9	2	6	9	6,930	23,975,280	0.000
9	4	4	65	712,530	23,975,280	0.030
10	3	6	28	123,032	60,141,120	0.002
10	5	4	126	3,945,060	60,141,120	0.066
10	7	2	344	27,019,824	60,141,120	0.449
11	4	6	65	1,067,430	148,729,680	0.007
11	8	2	513	76,920,246	148,729,680	0.517
12	5	6	126	5,913,684	325,832,976	0.018
12	7	4	344	54,000,432	325,832,976	0.166
12	9	2	730	194,108,460	325,832,976	0.596
13	8	4	513	153,765,594	677,714,400	0.227
14	7	6	344	80,981,040	1,326,627,120	0.061
14	9	4	730	388,084,060	1,326,627,120	0.293
14	11	2	1,332	944,241,480	1,326,627,120	0.712
15	8	6	513	230,610,942	2,467,618,800	0.093
16	9	6	730	582,059,660	4,220,872,800	0.138
16	11	4	1,332	1,888,128,648	4,220,872,800	0.447
16	13	2	2,198	3,537,245,796	4,220,872,800	0.838
18	11	6	1,332	2,832,015,816	11,787,972,672	0.240
18	13	4	2,198	7,073,687,124	11,787,972,672	0.600
19	16	2	4,097	18,331,903,590	19,233,722,496	0.953
20	13	6	2,198	10,610,128,452	30,518,778,360	0.348
20	17	2	4,914	29,656,019,484	30,518,778,360	0.972

None of these beats the lab's Haar graphs; the closest are $(19, 10)$ ($q = 16$, 95%) and $(20, 10)$ ($q = 17$, 97%). The reason is visible in the count: the 3-balls of lines that Lemma 4.3 uses have $(q+1)(q^2+1)$ elements, but a class has to meet all of them and the spread stars have q^2+q+1 lines, so $K \le (q+1)(q^2+1)$ while Corollary 4.4 has $K = q^3+1$; and the sign trick needs two matchings per factor. **Heuristic 4.6:** a partition of the lines into $K \approx (q+1)(q^2+1)$ classes meeting every set $\{l : d(p,l) \le 3\}$ would roughly add a factor $1 + 1/q$; it would not change the conclusion at $D = 10$.

5. Diameter 11: disjoint lines and non-collinear points

Lemma 5.1 (Theorem). In $H(q)$, (a) the point set of any line meets $B_4(p)$ for every point p; (b) the pencil of any point z (the $q+1$ lines through z) meets $B_4(l)$ for every line l.

Proof. (a) $d(p,\ell) \in \{1,3,5\}$; the point of ℓ on the (unique, by (F2)) shortest path from p to ℓ is at distance $d(p,\ell) - 1 \le 4$ from p. (b) is the dual statement: the line through z on the shortest path from l to z is at distance $d(l,z) - 1 \le 4$ from l. □

Corollary 5.2 (Theorem). Suppose $H(q)$ has K pairwise disjoint lines $\ell_0, \dots, \ell_{K-1}$ and K pairwise non-collinear points $z_0, \dots, z_{K-1}$. Then for every $r \geq 1$ there is a bipartite $(q+1+r)$-regular graph of diameter at most 11 with $2N(q)(2rK+1)$ vertices.

Proof. Let A_i be the point set of ℓ_i and B_i the pencil of z_i ($|A_i| = |B_i| = q+1$); they are pairwise disjoint and 4-dominating by Lemma 5.1. Distribute the remaining $N - K(q+1)$ points and lines evenly so that $|A_i| = |B_i|$ stays true (Remark 2.5) and apply Theorem 2.3 with $\rho = 4$, $c = 5$. □

Lemma 5.4 (Theorem). (a) Two points of $H(q)$ are at distance ≤ 4 in I if and only if they are orthogonal with respect to b. (b) Every totally singular line m of $Q(6,q)$ (a line of the quadric, not necessarily a line of H) contains a point at distance ≤ 4 from any given point p; so the $q+1$ points of m form a 4-dominating point set.

Proof. (a) Points at distance 2 lie on a common line of H, which is totally singular; points x, y at distance 4 have a common neighbour point z at distance 2 from both, so both lie in the totally singular plane π_z. In both cases $b(x,y) = 0$. Conversely, $p^\perp$ is the tangent hyperplane of $Q(6,q)$ at p (for q even, p is not the nucleus $\langle e_3 \rangle$ of b, so $p^\perp$ is still a hyperplane), and $p^\perp \cap Q(6,q)$ is a cone with vertex p over a parabolic quadric $Q(4,q)$: it has $1+q(q^3+q^2+q+1) = 1+q+q^2+q^3+q^4$ points. The points at distance ≤ 4 from p number $1+q(q+1)+q^3(q+1)$, the same; so the two sets coincide. (b) m is a 2-dimensional subspace and $p^\perp$ a hyperplane, so they share a nonzero vector, i.e. a point x of m with $b(p,x) = 0$; by (a), $d(p,x) \leq 4$. □

Corollary 5.5 (Theorem). Suppose $Q(6,q)$ has K pairwise disjoint totally singular lines (a partial line spread of the quadric) and $H(q)$ has K pairwise non-collinear points. Then the conclusion of Corollary 5.2 holds: for every $r \geq 1$ there is a bipartite $(q+1+r)$-regular graph of diameter at most 11 with $2N(q)(2rK+1)$ vertices. If q is a power of 3, the second hypothesis can be dropped: with a duality φ, the line classes $\varphi(m_i)$ (each a set of $q+1$ lines of H) are 4-dominating (Remark 2.6 and Lemma 5.4(b)), pairwise disjoint and balanced with the point classes.

Proof. As for Corollary 5.2, with Lemma 5.4(b) for the point classes. □

Computer-checked 5.6. $Q(6,3)$: a randomized greedy finds 78 pairwise disjoint totally singular lines (85.7% of the bound 91; qspread.py); with the duality of $H(3)$ (research/d10D7/dual/dualfind.c) both sides have 78 classes, and the independent scorer domcheck.py confirms that every class meets every 4-ball. The lift with $r = 1$ (lift9.py) has 114,296 vertices; apsp: bipartite, 5-regular, diameter exactly 11 (checks/d11_q3_K78_r1_D11.apsp.txt; the run with $D = 10$ fails). Squad card r1/c13 built the pencil version at $q = 2$ ($K = 19$, the largest partial distance-2 spread of $H(2)$, found by exhaustive search): diameter 11 for $r = 1$ and 10 for $r = 2$.

Pairwise disjoint lines are a partial distance-2 spread and pairwise non-collinear points a partial distance-2 ovoid; both have at most $q^4+q^2+1 = N/(q+1)$ elements (each point lies on at most one of the lines; each line contains at most one of the points). Distance-2 ovoids attaining the bound are known only for small q (§9). The sizes, and the fraction of the bound that K must reach to beat the lab's current values at $(9,11)$ and $(10,11)$:

Delta	q	r	max K	order at max K	lab value	K needed to beat it
4	2	1	21	5,418	19,152	impossible
5	2	2	21	10,710	253,080	impossible
5	3	1	91	133,224	253,080	impossible
6	2	3	21	16,002	1,971,744	impossible
6	3	2	91	265,720	1,971,744	impossible
6	4	1	273	1,493,310	1,971,744	impossible
7	2	4	21	21,294	10,648,176	impossible
7	3	3	91	398,216	10,648,176	impossible
7	4	2	273	2,983,890	10,648,176	impossible
7	5	1	651	10,179,036	10,648,176	impossible
8	2	5	21	26,586	43,702,560	impossible
8	3	4	91	530,712	43,702,560	impossible
8	4	3	273	4,474,470	43,702,560	impossible
8	5	2	651	20,350,260	43,702,560	impossible

Delta	q	r	max K	order at max K	lab value	K needed to beat it
9	2	6	21	31,878	145,258,344	impossible
9	3	5	91	663,208	145,258,344	impossible
9	4	4	273	5,965,050	145,258,344	impossible
9	5	3	651	30,521,484	145,258,344	impossible
9	7	1	2,451	192,276,048	145,258,344	1852 (76%)
10	2	7	21	37,170	445,709,568	impossible
10	3	6	91	795,704	445,709,568	impossible
10	4	5	273	7,455,630	445,709,568	impossible
10	5	4	651	40,692,708	445,709,568	impossible
10	7	2	2,451	384,512,880	445,709,568	impossible
10	8	1	4,161	623,376,054	445,709,568	2975 (71%)
11	9	1	6,643	1,765,310,820	-	-
12	9	2	6,643	3,530,488,780	-	-
13	11	1	14,763	10,461,770,424	-	-
14	11	2	14,763	20,923,186,536	-	-
15	13	1	28,731	46,227,144,684	-	-
16	13	2	28,731	92,453,484,900	-	-
17	13	3	28,731	138,679,825,116	-	-
18	16	1	65,793	294,355,118,694	-	-
19	16	2	65,793	588,708,000,426	-	-
20	17	2	83,811	1,011,499,873,020	-	-

(Rows with "-": no lab value; the row shown is the largest order at that degree, at the upper bound for K.)

Computer-checked 5.3 (squad cards r1/c11, r1/c12; sets re-checked against the hexagon files). $H(7)$ has 1,619 pairwise disjoint lines and 1,618 pairwise non-collinear points (iterated local search, research/gh6/ils_mis.c), so $K \geq 1{,}618$ at $q = 7$ (66% of the bound; 1,852 would be needed at $(9, 11)$). $H(8)$ has 2,701 pairwise disjoint lines and 2,701 pairwise non-collinear points, each a union of 37 orbits of the Singer subgroup of order 73 of $SL(3, 8)$ (local search on the orbit quotient), so $K \geq 2{,}701$ at $q = 8$ (65%; 2,975 would be needed at $(10, 11)$).

For $\Delta \geq 11$ the lab has no values at $D = 11$ (the 2013 table stops at $D = 10$, and the lab's beyond-table band covers $D = 11$ only for $\Delta \leq 10$).

6. Diameter 8, quadrangles and octagons (remarks)

- $D = 8$**.** Theorem 4.2 with $\rho' = 1$ asks for classes of lines meeting every pencil, i.e. sets of lines covering every point; there are at most $q + 1$ disjoint ones, so $m \leq 2r'(q + 1) + 1$ layers at degree $q + 1 + 2r'$. This is the regime of the lab's diameter-8 compound note (research/theorems/D8_COMPOUND.md, branch `claude/own-d8-compound-1z4e6s`: about $2q$ layers), and below the 2013 table's pattern at $D = 8$. No new cell.
- **Quadrangles.** For I the incidence graph of a generalized quadrangle of order (q, q) ($c = 3$), Theorem 2.3 gives diameter $5 + \rho$. At $\rho = 2$ ($D = 7$) a class must meet every $p^{\perp}$ ($q^2 + q + 1$ points), so $k \leq q^2 + q + 1$ and the order is at most $2(q + 1)(q^2 + 1)(2r(q^2 + q + 1) + 1)$. For every $5 \leq \Delta \leq 20$ this bound is below the plain hexagon level graph $(2r + 1) \cdot 2N(q)$ of the same degree (checked for all prime powers $q \leq 19$; at $\Delta = 4$ it is 450 against the lab's 864). Mixing hexagon and quadrangle layers does not help in this framework: the bound $\rho + c + 2$ is set by the largest point–line distance c.
- **The Ree–Tits octagon** $GO(q, q^2)$, $q = 2^{2e+1}$, has incidence graph of diameter 8 with point degree $q^2 + 1$ and line degree $q + 1$ and $|L|/|P| = (q^2 + 1)/(q + 1)$. Balancing the degrees by duplicating points gives maximum degree $\geq q^2 + 1$: $\Delta = 5$ for $q = 2$ (the incidence graph itself, 4,680 vertices, diameter 8, against the table's 9,234 at $(5, 8)$) and $\Delta \geq 65$ for $q = 8$. Neither falls in a cell the lab tracks ($\Delta \leq 20$), so octagons give nothing here.

7. Where this leaves diameters 8–10

- $D = 8$**.** No construction here reaches the lab's values or the 2013 table. The table's $D = 8$ entries are Gómez's compounds ΩH_q, which the lab found not to have diameter 8 as published (papers/Ozden_OmegaH_D8_note.pdf); the lab's own compound note gets about $2q$ layers, and §6 gives at most $2r'(q+1)+1$.
- $D = 9$**.** Corollary 3.1 is the first construction in this family whose order grows like $4rq^2 \cdot q^5$: with k at its maximum q^2+q+1 it would beat the lab's value at $D = 9$ for every $7 \le \Delta \le 20$ except $\Delta = 8$ (table in §3). The obstacle is the size of k that can actually be realised (Computer-checked 3.3, 3.4): perfect partitions do not exist at $q = 3$, Hermitian spreads of $H(9)$ pairwise intersect (Computer-checked 7.1), and the local searches reached only about half of the maximum. The record thresholds need $k/(q^2+q+1)$ between 49% ($(20, 9)$, $q = 17$) and 97%.
- $D = 10$**.** Corollary 4.4 holds for every q with no search at all; it reaches 84–97% of the lab's Haar values at $(16, 10)$, $(19, 10)$, $(20, 10)$ and beats none.
- $\Delta \ge 17$**.** The beyond-table cells at $D = 8$–10 hold Haar graphs at 4.5–7% of the Moore bound; the ball lifts reach that range only at $D = 9$ with large k (e.g. $(19, 9)$ and $(20, 9)$ need $k \ge 136$ at $q = 16$, half the maximum 273).

Computer-checked 7.1 (squad card r1/c02, research/squads/polylev/r1/c02/). In $H(9)$, 5,000 Hermitian spreads (lines of $H(9)$ in a hyperplane $v^\perp$ with q^3+1 lines of H) were sampled; every one of the 12,497,500 pairs meets, in 1 line (22.0%) or 10 lines (78.0%). One fixed Hermitian spread was compared with the line sets of all 597,871 hyperplanes of $PG(6, 9)$: the intersections have 1, 10 or 730 lines, never 0. Since $G_2(9)$ acts transitively on the elliptic hyperplanes (**Heuristic**: we did not check this transitivity by computer), no two Hermitian spreads of $H(9)$ are disjoint, and by Remark 2.6 the same holds for the Hermitian ovoids.

8. Computer checks

All files are under research/theorems/polylev/ unless stated otherwise; `hexgen2` is research/d10D7/hexgen2.c (the lab's generator of $H(q)$, which checks N points, N lines, $q+1$ points per line and lines per point), and `apsp` is research/lifts/coordinator_check/apsp.c (all-pairs bit-parallel BFS: simple, maximum degree, bipartite, connected, exact diameter; it assumes no symmetry).

claim	check	result
Theorem 2.3 / Cor. 3.1, $q = 3$, $k = 7$	lift9.py + apsp, $r = 1, 2, 3$ (10,920; 21,112; 31,304 vertices)	bipartite, regular of degree $q+1+r$; diameter 9, 9, 8; "$D = 8$" fails for $r = 1, 2$
Cor. 3.1, $q = 4$, $k = 13$	squad r1/c13: translocal colourings, lift9.py, apsp (73,710; 144,690 vertices)	diameter ≤ 9 for $r = 1, 2$; "$D = 8$" fails
Cor. 4.4, $q = 2, 3$, $r' = 1$	squad r1/c14: build_star.py + apsp (2,394; 41,496 vertices, 2 seeds each)	diameter 8 ($q = 2$), 9 ($q = 3$)
Cor. 5.2, 5.5	$q = 3$: 78 singular-line classes + duality, lift9.py + apsp (114,296 vertices); squad r1/c13: $q = 2$ pencils, $K = 19$	diameter 11 (and 10 fails); $q = 2$: diameter 11 ($r = 1$), 10 ($r = 2$)
Comp. 3.3	perfcodes.c (exact cover), triples.py	28 spreads, no ovoid in $H(2)$; 6,318 spreads and 6,318 ovoids in $H(3)$; no 3 pairwise disjoint spreads
Comp. 3.4	domlocal.c, translocal.c; independent scorer domcheck.py (tests every 2-ball by BFS)	see the table in §3

claim	check	result
Comp. 3.6	g2.py (collineations by apartment maps, verified on all lines), singer_perms.py (orbit sizes printed)	semiregular groups listed in §3.5
Comp. 5.3	squad r1/c11, r1/c12 (ils_mis.c, mis2.c; sets re-verified)	$K \geq 1{,}618$ ($q = 7$), $K \geq 2{,}701$ ($q = 8$)
Comp. 7.1	squad r1/c02 (sample.py, full_check.py)	Hermitian spreads of $H(9)$ pairwise meet

The squad runs, their scoreboard and their logs are in research/squads/polylev/r1/ (log of the rounds in research/squads/LOG.md).

9. Prior art

Searches by squad card r1/c16 (research/squads/polylev/r1/c16/prior_art.md, with evidence tags) and the lab's novelty file (research/lit/novelty-2026-10-25.md on branch `claude/novelty-literature-check-4i5frp`).

- **Level graphs ($\rho = 0$).** The $D = n + 1$ level graphs of generalized polygons with a tournament of layers are the Gómez–Fiol compounds $B(T)$ (J. Gómez, M. A. Fiol, *Dense compound graphs*, Ars Combin. 20-A (1985)) and give the 2013 table's diameter-7 column, credited to Delorme, Gómez and Quisquater (*On large bipartite graphs*, cited as submitted; not found). Remark 2.4 is that construction.
- **Voltage lifts of polygon incidence graphs at larger diameter.** C. Delorme, *Grands graphes de degré et diamètre donnés*, European J. Combin. 6 (1985) 291–302, §4: products $G * \Gamma'$ of the incidence graph G of a generalized quadrangle or hexagon with an oriented perfect matching; item (c), $G * C_{35}$, has degree $\Delta + 2$, diameter $D + 3$ and $35n$ vertices, i.e. for $H(q)$ degree $q + 3$, diameter 9 and $70N(q)$ vertices (card c16 read the full text; bipartiteness of (c) is not stated there; the bipartite versions are in C. Delorme, *Large bipartite graphs with given degree and diameter*, J. Graph Theory 9 (1985) 325–334, not read). This is the closest prior construction to Corollary 3.1, which has $2N(4k + 1)$ vertices at the same degree and diameter, more than $70N$ as soon as $k \geq 9$. Delorme's layers are indexed by a fixed cyclic group and every jump has a fixed voltage per arc; the ball lift lets the voltage depend on the colour class of the jumping element, which is what makes the number of layers grow like q^2.
- **Hubs and freed sets** (D7 note; R. Rajiv, Zenodo 2026, doi 10.5281/zenodo.23052995) concern diameter n and $n + 1$; not used here.
- **Ovoids, spreads, partitions.** Thas (J. Combin. Theory Ser. A 29 (1980)) relates ovoids of $H(q)$ to perfect codes (not read). Distance-2 ovoids of $H(q)$ are known only for $q = 2, 3, 4$ (De Wispelaere–Van Maldeghem 2008, cited by Bishnoi–Ihringer, arXiv:1606.07288). For partitions of the points of $H(q)$ into ovoids, of its lines into spreads, or the domatic number of its collinearity graph: nothing found in these sources.
- **Bipartite values at $D = 9$, $D = 11$ after 2013:** nothing found in these sources (the CombinatoricsWiki page could not be retrieved on 28 October 2026).